$$A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 3 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)^2 - 6$$

$$|A - \lambda I| = \boxed{\lambda^2 - 2\lambda - 5 = 0}$$

Definition :

Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n.

Consider the homogeneous system of equation

$$AX = \lambda X \text{ or } (A - \lambda I)X = O \qquad \text{..... (1)}$$

Where I is an indentity matrix of order n and λ is a scalar.

We know that homogeneous system of equation (1) always has a trivial solution. Here our concern is to find the values of λ, for which non-trivial (non-zero) solutions of the homogeneous system exist.

$$A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots\cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots\cdots & a_{2n} \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & \cdots\cdots \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & \cdots\cdots \\ a_{n_1} & a_{n_2} & \cdots\cdots & a_{nn} - \lambda \end{bmatrix}$$

is called a characteristic matrix of A, and the determinant

$$\det A = | A - \lambda I | = \begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots\cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots\cdots & a_{2n} \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & \cdots\cdots \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & \cdots\cdots \\ a_{n_1} & a_{n_2} & \cdots\cdots & a_{nn} - \lambda \end{bmatrix}$$

is called the characteristic polynomial of A.

Also, the equation $|A - \lambda I| = 0$ is called the characteristic polynomial of A.

Note :

1. In this chapter we are considering a homogeneous system as follows :

$$AX = \lambda X$$

$$AX - \lambda X = O$$

$$(A - \lambda I)X = O \qquad \text{.... (1)}$$

Non zero solution $X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$ of (1) is called eigen vector or characteristic vector,

Note :

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$A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 1 \\ 3 & 2-\lambda \end{vmatrix}$

$= \lambda^2 - (6)\lambda + 5$

$\lambda^2 - 6\lambda + 5 = 0$

$\lambda^2 - 5\lambda - \lambda + 5 = 0$

$\lambda(\lambda - 5) - 1(\lambda - 5)$

$\lambda = 1, 5$

$x = 1$

$\underbrace{(x-1)}_{=0}\underbrace{(x-5)}_{\neq 0}$

$\lambda = 1$

$(A - I)(X) = \begin{pmatrix} 3 & 1 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$R_2 \to R_2 - R_1$

$= \begin{pmatrix} 3 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$3x_1 + x_2 = 0$

$x_1 = k \Rightarrow x_2 = -3k$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = k \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$